# Examiner reports

## Q1.

This question also provided more challenge for the candidates. In this question a number of candidates did assume that the acceleration due to gravity was positive.

In part (a), quite a number of candidates found the time of flight and then used half of this value to calculate the maximum height. There were however a number of cases where the candidates did not half their value for the time of flight and so produced solutions which did not relate to the maximum height.

In part (b), as might be expected some candidates had difficulty with the signs in their equations. Those who were able to form a reasonable quadratic often went on to solve this to find a value for the time. However quite a few candidates did not show both solutions of the quadratic. It is important for candidates to show both solutions and indicate which one they have selected. Some candidates did not show enough intermediate working to gain full marks in this "show that" type question.

A few candidates used a method which involved finding the time taken to move from the highest point to the point where it was caught and adding this to the time to reach the maximum height. This method was often used by candidates who had found a time in part (a).

In part (c) a large number of candidates simply found the vertical component of the velocity when the ball was caught and did not consider the horizontal component at all.

## Q2.

This question was usually answered correctly, although some obtained $-4\cos 2t$ rather than $+4\cos 2t$. A common error was the omission of $+c$, while others added $+c$ but automatically assumed that $c = 0$.



## Q3.

Most of this question was answered well. In part (a), the only difficulty was found in differentiating $2e^{\frac{1}{2}t}$. In part (b), many found the velocity of the particle to be $(e^{\frac{3}{2}} - 8)\mathbf{i}$ [or $-3.52\mathbf{i}$], but did not find the speed of $+3.52$ m s$^{-1}$. Others gave the speed as 3.5 m s$^{-1}$, which was penalised. Those candidates who were successful in part (a) usually completed parts (c) and (d) correctly.

## Q4.

Part (a) of this question was done very well, but part (b) caused difficulties for some candidates. The most common errors in part (b) were to include the weight of the car and trailer or to produce equations in which the wrong combinations of forces were used.

## Q5.

Parts (a) and (b) were found to be fairly straightforward by candidates, but part (c) was much more demanding. The candidates often calculated the vertical component of the velocity in part (c), but were unable to use this correctly in further calculations. There was often confusion about how to calculate the speed and the direction of motion.

## Q6.

There were many very good responses to this question. Generally, the candidates who did have difficulties with this question had problems in part (a) but were able to complete part (b) and gain the follow-through marks by using their answer to part (a). The sort of difficulties encountered in part (a) included:

- using a value for the acceleration, for example $g$ or zero in their constant acceleration equation;
- simply using $u = \frac{s}{t}$ ;
- using $s = \frac{1}{2}(u+v)t$ , but then having difficulty solving for $u$.

## Q7.

The candidates found this question more demanding than the previous questions. In part (a), there were many good responses, but a number of candidates lost marks because they did not fully justify their final answer. When giving answers to questions like part (a), where an answer correct to three significant figures is required, candidates should be advised to write an answer to more than three significant figures before giving their final answer; otherwise examiners do not know whether the value has been calculated or copied from the question paper.

Part (b) was generally done well, sometimes using the answer given in part (a).

In part (c), there was quite a mixture of responses. A significant number of candidates thought that the distance would change. Of those who felt that the distance would not change, only some were able to justify this statement.



Part (d) was done well by some candidates, but one of the most common errors was to use a time of 3.13 seconds. A few candidates did not resolve the velocity and used 20 instead of 20 sin 50° in their calculations.

In part (e), candidates were simply expected to write down the velocity, but very few candidates did this. A significant number tried to calculate the velocity. In this case, there were often minor errors in the calculation of the speed. When stating the direction of the velocity, some candidates simply stated 50°, but did not specify that this was below the horizontal or draw a diagram to show this.

## Q8.

Some candidates found it very difficult to form a correct quadratic equation based on the vertical motion of the pellet. One of the main difficulties was dealing with the sign of the number 3.

Many candidates did part (b) well, and some who had been unable to find the time were able to use the printed answer to calculate the range. Parts (c) and (d) were found more challenging by candidates. While there were a few very good solutions, many candidates did not realise how to obtain the required answers, especially in part (d).

**Q9.**

Many candidates did well on this question, scoring a good number of marks. Part (a) was often answered well, but some candidates did include a number of spurious reasons. In part (b), there were many good responses. Some candidates did not show clearly how they obtained the equation which they went on to solve to find the time.

A large number of candidates found the time to the maximum height and doubled this value. Some of these candidates did not explain why this doubling had taken place and a 2 seemed to appear in their equations with no justification. Part (c) was generally done well. Some candidates gained full marks on part (c) without doing part (b). A few candidates had difficulty solving the inequalities to find the values of $V$.

**Q10.**

Generally the candidates made quite good attempts at this question, but some parts caused the candidates more difficulty. A few candidates made poor use of vector notation. Part (a) was answered correctly by the vast majority of the candidates. The most common error was dealing incorrectly with the initial velocity. A few went on to simplify their answers incorrectly. Part (b) was a little more demanding, as some candidates confused the components. Part (c) did cause a little more difficulty for some candidates, with a few candidates not giving any answer at all. However, there were again many correct solutions.

In part (d)(i), although there were many correct solutions, some candidates showed that the helicopter was south or north of the origin, rather than specifically north. Interestingly, some candidates who could not answer part (c) were able to write down the position vector at the specified time. A few candidates found the correct position vector but did not conclude that the helicopter was due north.

When answering part (d)(ii), many of the candidates found the correct velocity, but some did not find the speed as requested.



**Q11.**

There were a fair number of good attempts at parts (a) and (b) of this question, but part (c) was found to be more demanding. In part (a), there were three main sources of error: introducing a sign error; using cos instead of sin; and not applying the quadratic equation formula correctly.

There were many good answers to part (b), with some candidates using the time given in part (a) when they had been unable to obtain it for themselves.

There were very few complete answers to part (c). Some candidates did not make any form of serious attempt at this part of the question. Some candidates only found one component of the velocity, but not the other; most often they found the vertical component, but not the horizontal component.

**Q12.**

The first two parts of this question were done very well by the vast majority of candidates, but part (c) caused more difficulties. In part (a), the candidates almost always drew a graph with the correct shape, but made errors on the axes. The two most common errors were to use 21 instead of 12 on the time axis and to use an incorrect value on the velocity axis. In part (b), a very small number of candidates made arithmetic errors and a few also used incorrect values.

Part (c) caused more difficulties for the candidates. There were three main incorrect approaches. The first of these was to assume that the forces on the lift were in equilibrium and hence simply state that the tension was equal to the weight, giving their answer as 2940 N. Some candidates used an incorrect value for the acceleration, which they then included in a three term equation of motion which would have otherwise been correct. The most common approach was to take the acceleration as 2 m $s^{-2}$.

A few candidates used a three term equation of motion with the correct terms but an incorrect sign. This approach would produce a result such as $T = 2940 - 150 = 2790$ N.

**Q13.**

Part (a) was usually done very well, but in some cases very badly. Many candidates produced good complete solutions. A few made arithmetic or algebraic errors that often resulted in the loss of the final accuracy mark. A few candidates found the time, but not the height. Part (b) was more demanding. Some candidates did not really make any progress with this part of the question. For those that got started, the most common errors were to include a sign error in the quadratic equation that they used to find the time; and to produce a correct quadratic equation, but to make an error in finding its solutions.

Some candidates did use a two-time approach, finding the time to go up to the maximum height and the time to come down to the tree. Many of these candidates were successful, but there were problems associated with getting the right distance to use to find the time down to the tree.

**Q14.**

While candidates realised that part (a) required differentiation, a significant number were not able to carry this out accurately. Part (b) proved more successful, but only a small minority of candidates were able to attempt part (c), the simplest solution involving maximisation of the appropriate trigonometrical function.

**Q15.**

In part (a) the majority of the candidates gained full marks, although there were a small number of candidates with poor responses, such as “has no mass”. Parts (b) and (c) were also done very well, but with some minor errors present. There were fewer correct responses to part (d).

In a number of cases the speed was given correctly and the angle of 40° was also stated, but no indication that it was below the horizontal was included. A few candidates worked out the horizontal and vertical components of the velocity when the arrow hit the ground and then calculated the speed and the direction. This was a very hard way to earn the marks, but it did work for some of these candidates. Generally, part (d) was either done well or very badly with candidates not knowing how to start.

**Q16.**

Parts (a), (b) and (c) of this question were often done very well. In part (a) candidates occasionally gave the answers “5” or “5**j**”, but the vast majority of the answers were correct. In part (b), many correctly stated the velocity, but some candidates then performed incorrect simplifications. Part (c) was also done well by those candidates who had obtained the velocity vector.

Part (d) proved to be more challenging. Some candidates realised that they needed to find the position vector and hence the bearing. These candidates usually made good progress, although some stopped when they had the position vector and others gave the answer as 52.3°. Some candidates tried to work with velocities.

## Q17.

This question proved difficult, and a number of candidates showed a lack of appreciation of the differences between the horizontal and vertical components of the motion. Solutions in part (a) sometimes lacked convincing reasoning. Algebra let some down in part (c), and part (d) proved very discriminating on the understanding of the motion.

## Q18.

Part (a) was done well, but in part (b) many stopped after finding the velocity at the given time, instead of continuing to find the speed by evaluating the magnitude of the velocity vector.

## Q19.

A straightforward question, but not always yielding the marks it should. In part (a) many were careless with working, often confusing values of $u$ and v, and altering or ignoring incorrect signs in subsequent working. Candidates should be aware that ‘show that’ requests require convincing correct working leading to the printed result. In part (b) sketches were usually good but sometimes marred by incomplete labelling. Part (c) was done well. Responses in part (d) sometimes referred to irrelevant factors rather than the model of the motion given in the question.



## Q20.

This was found to be the most difficult question on the paper. While the most able were able to score full marks, others showed confusion between vectors and scalars, or simply showed misunderstanding of vector quantities. Many found the vector $\vec{AB}$ in part (a) but could not then proceed correctly, with ratios of vectors often leading to vector expressions for the time. The most successful methods included the application of the equations of motion with constant acceleration, and the use of diagrams with explained steps in motion. Part (b) often began with the assumption of a zero initial velocity, some only considered the **j** component of the motion, and many failed to add the position vector of the particle at B. Part (c) often began well with the vector $\overrightarrow{AC}$ found, but with no attempt to find its magnitude to give the distance requested.

## Q21.

This question proved quite challenging, with the least successful candidates confusing the horizontal and vertical components of motion, and in some cases trying to incorporate expressions for the greatest height, horizontal range, and time of flight, thereby making the question harder. Part (a) was mostly done well, but some assumed vertical motion was involved. Part (b) was sometimes omitted, while a significant number did not realise that the calculation could be done in one stage and wasted time in carrying out a series of calculations often losing an accuracy mark due to rounding values within their working. In part (c)(i) common errors included finding the vertical component of velocity only, using '24' as the initial velocity with no component, and mixing horizontal/vertical motions or velocity/displacement in calculations. Those successful in part (c)(i) usually completed part (c)(ii).

## Q22.

A popular question with high marks awarded. It was pleasing that many found the average speed as requested although some found the final speed instead.

## Q23.

A straightforward question, but not always yielding the marks it should. In part (a) many candidates were careless with working, often confusing values of $u$ and v, and altering or ignoring incorrect signs in subsequent working. Candidates should be aware that 'show that' requests require convincing correct working leading to the printed result. In part (b) sketches were usually good but sometimes marred by incomplete labelling. Part (c) was done well. Responses in part (d) sometimes referred to irrelevant factors rather than the model of the motion given in the question.

## Q24.

There were very many good solutions to this question and a good number of candidates scored full marks. Some candidates did however make errors. In part (a) there were some cases where the candidates made some minor errors when differentiating one or both of the components. In part (b) a few candidates had difficulties substituting $t$ = 1/3 correctly, but the biggest problem in this part of the question was the description of the direction. While a number did state that the direction was south, some stated that it was north, even though they had calculated the velocity correctly. Others did not use either north or south, but gave answers that implied that **j** was a vertical unit vector.

Parts (c) and (d) were generally done well, but a few candidates experienced difficulties with the differentiation and very occasionally with the substitution of $t$ = 4 .

## Q25.

Most candidates did well on this question; almost all were able to obtain the acceleration without difficulty. Finding the distance was also done well, but a few candidates made arithmetic errors. There was also a small number who assumed a constant velocity. Part (b) caused a few more problems. Some candidates had trouble with the units and did not convert tonnes to kg. A few also made sign errors, while some others simply calculated the product of the mass and acceleration.

## Q26.

Part (a) of this question was done well by several candidates, while part (b) was found to be more demanding. In part (a), the candidates seemed to have been helped by the printed answer. In part (b), the major problem was setting up an appropriate equation to find the time of flight. Those who could do this generally went on to complete the question correctly. One issue that was evident was that a small number of candidates did not resolve the velocity into components, but simply worked with 30 wherever they needed a velocity.

## Q27.

Candidates found this question quite demanding. Candidates were most successful with part (a) and for some this was the only place where they gained marks. In part (b), many candidates omitted the initial position of the yacht, and a number tried to integrate the initial velocity. In part (c)(i), a poor answer for the position vector had an impact. However, some candidates who had good expressions simply showed that the i component of the position was zero and not that the yacht was due north of the origin. In part (c)(ii), quite a few candidates found the velocity correctly, but did not go on to find the speed.

## Q28.

Part (a) was generally answered well. There were however some curious assumptions, for example "the ball has no mass", which were not appropriate. Part (b) was generally done quite well, with candidates obtaining the results without too much difficulty. Part (c) proved to be more demanding, but candidates often tended to do very badly or very well. If they could produce a correct equation for $t$ they could often gain all the marks, but other candidates could not see how to form an equation that would enable them to start the question. In a few cases the candidates rounded the time of flight and then obtained an incorrect value for the range.

## Q29.

Part (a) was often done well, but a number of candidates effectively repeated the same assumption twice. The candidates were often helped by the printed answer in part (b). Many of the candidates made life difficult for themselves by finding the initial speed and angle of projection and then working with these values. One consequence of this was that the final answers were inaccurate. Part (c) proved more challenging for candidates. The most common incorrect approach was to assume that the ball hit the wall when it was at its maximum height. In part (d) there were some good answers and many of the candidates were on the right track. However quite of few of the candidates produced answers which lacked clarity of explanation.



## Q30.

This was done well by most, although there was occasional confusion seen in using the appropriate distance for parts (b) and (c).

## Q31.

Diagrams were mostly very good with only occasional lapses in the direction of the normal reaction force or in poor labelling of forces. Part (a) proved popular and successful for most candidates. In part (b)(i) a number failed to appreciate that the forces involved included the weight component, but many went on to be successful in part (ii).

## Q32.

This question proved more discriminating than expected.

**Q33.**

This question proved both popular and successful for candidates. Part (a) was very well done and part (b) only slightly less so with occasional accuracy errors or misunderstanding regarding the average speed.

**Q34.**

This question proved to be very demanding. Many candidates failed to understand the motion of the particle, not appreciating that its deceleration would cause the direction of motion to eventually reverse. There were consequently many erroneous attempts in part (a). Candidates who had found two velocities in part (a) were then able to gain marks in part (b), provided their values were consistent with the nature of the motion. There were, however, many elegant and concise solutions to both parts of the question.

**Q35.**

Part (a) was mostly completed successfully, showing a pleasing improvement in algebraic skills. Part (b) proved more discriminating. Many candidates found sketch graphs helpful in understanding the problem. The most direct method was the use of $s = \frac{1}{2}(u+v)t$, but many were successful with longer methods involving the retardation.